Econ 3040 - Assignment 2: Sales from different video game consoles

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The big three 7th generation video game consoles released in 2005-2006 were the Nintendo Wii, Playstation 3, and Xbox 360. Use the data below to determine if the critic score of a video game is related to its sales. The data for your assignment is the same as from Computer Lab 1. The data was obtained by Abdulshaheed Alqunber.

Due date: October 29th, 2026. Worth 2% of your final grade.

Instructions: Submit your assignment in the "Assignment 2" drop box on UM Learn. *You must complete your assignment individually.* Submit both your answers to the assignment questions, and the R code that you use.

Load the data using:

```
mydata <- read.csv("https://rtgodwin.com/data/vidsales.csv")
```

Make a subset that contains only `Wii`, `PS3`, and `X360` games using:

```
subdat <- subset(mydata, Platform == "Wii" | Platform == "PS3" | Platform == "X360")
```

Use this subset for the remainder of the assignment.

1. How much does it pay to make a *good* game? Estimate a model by least squares (LS) to answer the question: how much more does a video game sell on average, when it has a higher critic score rating?
2. Interpret the R-square from the estimated model in question 1.
3. Test the hypothesis that `Score` has no effect on `Sales`.
4. Estimate a new model using `Sales` as the dependent variable, and `Score` and `Platform` as the regressors (explanatory variables). What is the average difference in the sales for these three video game consoles?
5. In the above question, there are **three** categories for `Platform`. Why did R only use **two** *dummy* variables in the regression?
6. In the new model (with multiple regressors), which R-squared should you use, and why?
7. Estimate a model with `Sales` as the dependent variable, and `Score`, `Publisher`, `Platform`, and `Genre` as the regressors. (Do not include the entire table of output results in your assignment, it is too big). What is the effect of `Score` on `Sales`? What is the reason for including these additional variables in the model, when we only care about the effect of `Score`?